

Entity Relationship Diagram (ERD) Specification

Overview

This specification describes the OLTP Entity Relationship Diagram for the greenfield_mp project. The diagram is stored in `docs/diagrams/oltp_er_diagram.mmd` and renders on GitHub using Mermaid syntax.

Diagram Structure

Entities and Relationships

1. DEPARTMENTS Table

- **Primary Key:** `department_id` (INT, AUTO_INCREMENT)
- **Unique Key:** `department_name` (VARCHAR(60))
- **Relationships:**
 - One-to-Many with EMPLOYEES (current department)
 - One-to-Many with EMPLOYEE_JOB_HISTORY (in department)
 - One-to-Many with PROJECTS (owns)
 - Many-to-Many with PROJECT_ASSIGNMENTS (staffed by)

2. JOB_ROLES Table

- **Primary Key:** `job_role_id` (INT, AUTO_INCREMENT)
- **Unique Key:** `job_title` (VARCHAR(60))
- **Relationships:**
 - One-to-Many with EMPLOYEES (current role)
 - One-to-Many with EMPLOYEE_JOB_HISTORY (as role)

3. EMPLOYEES Table

- **Primary Key:** `employee_id` (INT, AUTO_INCREMENT)
- **Unique Key:** `email` (VARCHAR(120))
- **Attributes:**
 - `first_name` (VARCHAR(60))
 - `last_name` (VARCHAR(60))
 - `email` (VARCHAR(120), UNIQUE)
 - `hire_date` (DATE)
 - `department_id` (INT, FK → departments)

- `job_role_id` (INT, FK → `job_roles`)
- `job_level` (TINYINT)
- `monthly_income` (INT)
- `attrition` (TINYINT(1))
- **Relationships:**
- One-to-Many with `EMPLOYEE_JOB_HISTORY` (has versions)
- One-to-Many with `PROJECT_ASSIGNMENTS` (works on)
- One-to-Many with `PERFORMANCE_REVIEWS` (receives)

4. `EMPLOYEE_JOB_HISTORY` Table

- **Primary Key:** `history_id` (INT, AUTO_INCREMENT)
- **Attributes:**
- `employee_id` (INT, FK → `employees`)
- `department_id` (INT, FK → `departments`)
- `job_role_id` (INT, FK → `job_roles`)
- `job_level` (TINYINT)
- `monthly_income` (INT)
- `effective_start` (DATE)
- `effective_end` (DATE, NULL for current)
- `change_reason` (VARCHAR(40))
- **Relationships:**
- Many-to-One with `EMPLOYEES` (employee versions)
- Many-to-One with `DEPARTMENTS` (in department)
- Many-to-One with `JOB_ROLES` (as role)

5. `PROJECTS` Table

- **Primary Key:** `project_id` (INT, AUTO_INCREMENT)
- **Unique Key:** `project_name` (VARCHAR(100))
- **Attributes:**
- `department_id` (INT, FK → `departments`)
- `status` (VARCHAR(20), default 'Active')
- `priority` (VARCHAR(10), default 'Medium')
- `budget` (DECIMAL(12,2))
- `start_date` (DATE)
- `end_date` (DATE, NULL for ongoing)
- **Relationships:**
- One-to-Many with `PROJECT_ASSIGNMENTS` (staffed by)
- One-to-Many with `PERFORMANCE_REVIEWS` (reviewed on)

6. `PROJECT_ASSIGNMENTS` Table

- **Primary Key:** `assignment_id` (INT, AUTO_INCREMENT)

- **Attributes:**
- `employee_id` (INT, FK → employees)
- `project_id` (INT, FK → projects)
- `role_on_project` (VARCHAR(20), default 'Contributor')
- `allocation_pct` (TINYINT UNSIGNED, 1-100)
- `start_date` (DATE)
- `end_date` (DATE, NULL for ongoing)
- **Unique Constraint:** (employee_id, project_id)
- **Relationships:**
- Many-to-One with EMPLOYEES (works on)
- Many-to-One with PROJECTS (staffed by)

7. PERFORMANCE_REVIEWS Table

- **Primary Key:** `review_id` (INT, AUTO_INCREMENT)
- **Attributes:**
- `employee_id` (INT, FK → employees)
- `project_id` (INT, FK → projects, NULLable)
- `review_date` (DATE)
- `performance_rating` (TINYINT, 1-4)
- `job_satisfaction` (TINYINT, 1-4)
- `environment_satisfaction` (TINYINT, 1-4)
- `work_life_balance` (TINYINT, 1-4)
- `percent_salary_hike` (TINYINT, default 0)
- `review_score` (DECIMAL(5,2))
- **Unique Constraint:** (employee_id, review_date)
- **Relationships:**
- Many-to-One with EMPLOYEES (receives)
- Many-to-One with PROJECTS (reviewed on)

Draw.io Editing Instructions

To Edit in draw.io:

1. Open draw.io (diagrams.net)
2. Use **Arrange > Insert > Advanced > Mermaid** to insert the Mermaid diagram
3. Paste the Mermaid code from `docs/diagrams/oltp_er_diagram.mmd`
4. The diagram will render automatically
5. To modify, edit the Mermaid code directly
6. Use **Arrange > Layout > Horizontal/Vertical** to adjust layout if needed
7. Save the file and it will update on GitHub

Mermaid Syntax Notes

- `||--o{` denotes one-to-many relationship (one side, many other side)
- `|o--o{` denotes many-to-many relationship
- `{ TableName { ... } }` defines table columns and constraints
- `PK` = Primary Key, `UK` = Unique Key, `FK` = Foreign Key

Current Diagram (oltp_er_diagram.mmd)

```
erDiagram
    DEPARTMENTS ||--o{ EMPLOYEES : "current department"
    JOB_ROLES ||--o{ EMPLOYEES : "current role"
    EMPLOYEES ||--o{ EMPLOYEE_JOB_HISTORY : "has versions"
    DEPARTMENTS ||--o{ EMPLOYEE_JOB_HISTORY : "in"
    JOB_ROLES ||--o{ EMPLOYEE_JOB_HISTORY : "as"
    DEPARTMENTS ||--o{ PROJECTS : owns
    EMPLOYEES ||--o{ PROJECT_ASSIGNMENTS : "works on"
    PROJECTS ||--o{ PROJECT_ASSIGNMENTS : "staffed by"
    EMPLOYEES ||--o{ PERFORMANCE_REVIEWS : receives
    PROJECTS |o--o{ PERFORMANCE_REVIEWS : "reviewed on"
```